Venkata Lakshmi Parimala Pasupuleti

AI Model Risk Associate · AI/ML Validation · Accuracy · Explainability · Algorithmic Robustness · LLM Evaluation · Python

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PROFESSIONAL SUMMARY

AI/ML engineer with 7+ years production experience, now specializing in **AI model validation** at Intuit — independently validating financial analytics AI for **accuracy, reliability, and compliance with regulatory requirements**. Built AgentProbe, an open-source validation toolkit that **develops and implements validation methodologies and benchmark models** tailored to the specific characteristics of AI/ML systems — scoring outputs across 5 LLM providers on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance via 3 LangGraph agents. LeaseI is an 8-agent LangGraph platform where I designed the **validation plan and benchmarking models** achieving **93.6% precision, 87.9% F1** across 1,240 documents — **conducting thorough testing, identifying model risks and limitations, and documenting findings** for stakeholder review. Strong stack: Python, LangGraph, LangChain, scikit-learn, XGBoost, HuggingFace, OpenAI APIs, GCP/Vertex AI, AWS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI Model Validation: AI/ML Model Validation, Accuracy Assessment, Explainability, Algorithmic Robustness, LLM Evaluation, LLM-as-Judge, Validation Methodology Design, Benchmark Models, Sensitivity Analysis, Back-testing, Performance Monitoring, Model Risk Documentation, Regulatory Compliance Review

Python & ML: Python (9+ yrs), LangGraph, LangChain, FastAPI, scikit-learn, XGBoost, PyTorch, Pandas, NumPy, HuggingFace, spaCy, ChromaDB, Pinecone

Cloud & AI Tools: GCP/Vertex AI, AWS (S3, EC2, RDS, SageMaker), Azure OpenAI, OpenAI APIs, Groq, Ollama, Docker, GitHub Actions CI/CD, Git

Data & Analytics: SQL, Snowflake, PostgreSQL, SQLite, ETL Pipelines, Statistical Modeling, Regression Analysis, Feature Engineering, Data Quality Monitoring

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — AI Model Validation & Risk · Financial Analytics

- **Independently validate AI/ML models** for financial analytics — assessing **accuracy, reliability, and compliance**; develop and execute validation methodologies (LLM-as-judge + golden datasets + regression testing) tailored to the specific characteristics of each model; **conduct thorough testing of model outputs, identifying and documenting risks and limitations**; compile model validation reports with structured audit trails for business stakeholders and leadership review.
- Build end-to-end ML pipelines (scikit-learn, XGBoost) on Snowflake and AWS from data ingestion through feature engineering, model evaluation, and prediction APIs; integrate Anthropic Claude into Python/FastAPI financial services; **collaborate with AI researchers and model developers** to address identified issues and improve model performance; ship via GitHub Actions CI/CD.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Analytic Software Engineer — Data Quality, Compliance & Risk Analytics

- Automated compliance analytics pipelines across 1M+ financial records; built data quality monitoring tools assessing data accuracy and flagging inconsistencies before downstream reporting; managed security mechanisms and **data access governance** via AWS S3 Libsodium-encrypted pipelines; **100% PHI compliance, zero violations** over 7+ years.
- Optimized **200+ SQL queries (30% performance improvement)**; Python automation eliminating **80% of manual workload**; Docker containerization (50% faster microservices); **120+ Agile sprints**.

PROJECTS

AgentProbe — AI Model Validation & Benchmark Toolkit (Open Source)

AI Validation · Benchmark

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- **Develops and implements validation methodologies and benchmark models** tailored to AI/ML systems: 5-metric evaluation engine (factual accuracy, hallucination rate, retrieval quality, completeness, compliance) across 5 LLM providers; 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix); CI regression harness with golden datasets; 15-endpoint FastAPI backend.

LeaseI — 8-Agent AI Validation Platform with LLM-as-Judge CI

AI Validation · LangGraph

Python · LangGraph (8-agent) · FastAPI · Hybrid RAG (TF-IDF + semantic) · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_raq

- Designed **validation plan and benchmark models** for an 8-agent LangGraph compliance platform (triage → retrieval → reasoning → validation → HITL escalation); **LLM-as-judge CI evaluation across 1,240 test cases: 93.6% precision, 87.9% F1**; thorough documentation of model risks, limitations, and findings; every determination traceable via full metadata and audit trail.

CrashClarity — Predictive Modeling & Risk Analysis (104K Records)

ML · Risk Analytics

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Random Forest + Logistic Regression on 104K+ records; **back-testing, sensitivity analysis, and performance monitoring**; **85%+ classification accuracy**; interactive dashboard translating model outputs to stakeholder insights.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ML pipelines, LLM evaluation, anomaly detection, data governance

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017